

Question:

What are the minimum and maximum possible values of Glasgow Coma Score?

Answer Choices:

- A. Minimum = 3, Maximum = 15
- B. Minimum = 0, Maximum = 13
- C. Minimum = 0, Maximum = 15
- D. Minimum = 3, Maximum = 18

Reasoning + Answer:

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Finding Reasoning Paths :

1 . **Glasgow Coma Scale (GCS)** â”” **Pupillary response (RIG)** â”” **Minimum GCS** : Evaluate the minimum score based on the severity of consciousness (e .g ., coma or deep sedation).

2 . **Glasgow Coma Scale (GCS)** â”” **Motor response (MFR)** â”” **Minimum GCS** : Analy ze the impact of a completely uncooperative or flaccid motor response on the score .

3 . **Glasgow Coma Scale (GCS)** â”” **Eye-opening response (ROR)** â”” **Minimum GCS** : Consider the role of eye-opening response in determining the minimum score .

4 . **Glasgow Coma Scale (GCS)** â”” **Definition of "no response "** â”” **Minimum GCS** : Explore the logical sequence of scoring "no response " as the lowest possible value .

Reasoning Process :

1 . **Understanding the Glasgow Coma Scale (GCS):**

The GCS is a clinical scale used to assess the level of consciousness of a patient . It is based on three components : eye-opening response (ROR), motor response (MFR), and verbal response (VOR). The scale ranges from 3 (minimally conscious) to 15 (fully alert).

2 . **Defining the minimum score:**

The lowest possible score occurs in cases where a patient is in a state of no response or deep sedation . In such cases :

- The **eye-opening response (ROR)** is "no response ," scoring 1 .
- The **motor response (MFR)** is "flaccid , unresponsive to stimuli ," scoring 5 .
- The **verbal response (VOR)** is "none ," scoring 1 .

Adding these values together :

$\backslash (1 \text{ (ROR) } + 5 \text{ (MFR) } + 1 \text{ (VOR) } = 7 \backslash)$.

However , this is incorrect because the motor response can only score up to 5 , not 10 .

Instead , the correct calculation is :

$\backslash (1 \text{ (ROR) } + 3 \text{ (MFR - incomplete responsiveness) } + 1 \text{ (VOR) } = 5 \backslash)$.

Therefore , the minimum GCS score is **5** , not 3 .

3 . **Re-evaluating the definition of "no response ":

A "no response " can be broken down into :

- **Coma (ROR : 2 , MFR : 2 , VOR : 1)** : This scores $\backslash (2 + 2 + 1 = 5 \backslash)$.
- **Deep sedation (ROR : 1 , MFR : 5 , VOR : 1)** : This scores $\backslash (1 + 5 + 1 = 7 \backslash)$.

The primary distinction is the scoring of the motor response (MFR), which can only drop to 2 in cases of significant brain injury or coma .

4 . **Maximum score:**

The maximum score of 15 is achieved when all three components are at their highest :

- **ROR : 5 (all eyes open)**
- **MFR : 5 (assisted to move extremities)**
- **VOR : 5 (calls out "pain ")**.

Conclusion :

The **minimum possible value of a Glasgow Coma Score is 5** , which corresponds to coma , where the motor response is not fully cooperative or the patient is unresponsive . The **maximum possible value is 15** , which corresponds to a minimally conscious patient with normal eye opening , normal motor response , and normal verbal response .

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<answer>

Minimum = 5 , Maximum = 15

</answer>